

(a) Relative tactic occurrence (%)

		APT attacker model				baseline attacker	in (b)	
		c_1	c_2	c_3	c_4			
SCAN_HOST	Reconnaissance	5	1	1	2	2	c_1	
ENUM_HOST	Lateral movement	9	8	5	14	25		
SCAN_PORT	Discovery	8	12	17	8	25		c_2
	Initial access	3	6	1	<1	23		
EXPLOIT_VULN	Execution	8	14	9	17			
	Privilege escalation	3	3	7	3			
BRUTE_FORCE	Credential access	10	3	1	10	16	c_3	
SCAN_NEIGHBOR	Command and control	17	14	11	19	9	c_4	
	Resource development	0	0	1	3	—	baseline attacker	
	Persistence	8	13	10	4			
	Stealth	14	16	16	10			
dwell-only	Defense impairment	1	—	—	5			
	Collection	7	<1	4	4			
	Exfiltration	6	—	4	—			
	Impact	<1	8	12	—			

(b) Attack path variation (APV)